

The Melody of the Mysterious Stones: A VR Mindfulness Game Using Sound Spatialization

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Figure 1: Start Screen of the Melody of the Mysterious Stones, 2022.

ABSTRACT

The Melody of Mysterious Stones is a VR meditation game that utilizes spatial audio technologies. One of the most common mindfulness exercises is to notice and observe five senses including the sense of sound. As a way of helping the players with focusing on their sense of sound, *the Melody of Mysterious Stones* makes use of spatialized sounds as game elements. Our play tests showed that game players enjoyed playing missions with spatial audio elements. Also, they reported that spatial audio helped them with focusing on their sense of sound and therefore felt more engaged in meditation materials.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Applied computing** → **Arts and humanities**.

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KEYWORDS

spatial audio, meditation, sound visualization, virtual reality, gamification

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1 INTRODUCTION

Generally, a story in a video game progresses mainly based on visual elements. However, there is no doubt that sounds are essential elements in video games as well. Sounds can be used to provide players with intriguing experiences [4]. Therefore, some sound-based games such as *Dark Echo* and *Lurking* have implemented mechanisms where players rely on sound to have the main story progress in the game [4]. Not many games, however, utilize such mechanisms in VR games. In this research, a sound-based mechanism where players mainly depend on spatial sounds is implemented in the form of a VR meditation game.

According to the study conducted by Reinecke et al., playing video games is more effective in improving mental health recovery than watching a non-interactive movie clip [14]. Therefore, a video game could be potentially used as a tool for mental exercise as long as materials are properly gamified. Many meditation games were

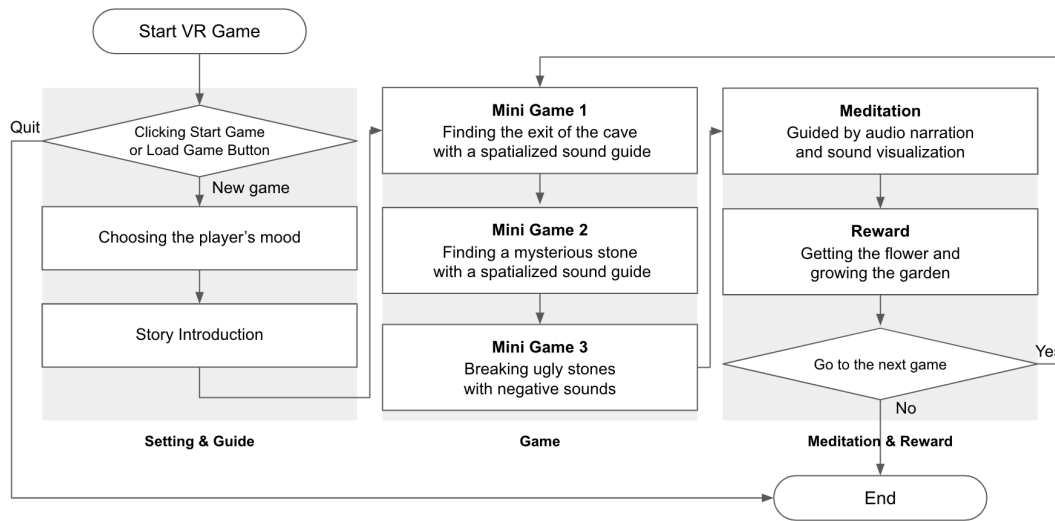


Figure 2: Flowchart of the game experience.

successful in attracting users with qualities of meditation materials and visual elements. In addition, a lot of studies have proven the efficacy of meditation games [19]. *Muse* is one of the meditation games that has shown positive results in reducing posttraumatic stress [15]. Especially, virtual reality mindfulness has been proven to be effective in increasing positive emotion without fostering negative emotions [16]. The steps of meditation games could be generally divided into two steps. One is the gameplay process that is separate from mediation exercises, and the other is meditation practice itself [20]. *The Melody of Mysterious Stones* is a VR game to make both steps intriguing to the user. Leveraging the effectiveness of spatial audio in representing the sense of direction and distance, this game increases the player's sense of immersion in VR experiences [11].

Because a sense of immersion is an essential element when providing qualities of meditation experiences, this game was designed as a VR game [2]. Previous works have proven that relaxing sounds are successful in relieving stress [1]. Accordingly, various games that make use of soothing sounds have been produced. *The Melody of Mysterious Stones* is also designed with an intention to provide players with a relaxing experience by making the most of calming soundtracks and audio guided meditation.

2 RELATED WORK

2.1 Sound-based Games

Some games have used sounds as the main elements of game play. For example, *Daydream* is an interactive sound-based adventure game [10]. It combines puzzles and interactive music-making experiences with the design concept of a “healing game” to help players slow down, relax, and imagine. In the preliminary user study that Ji and Nishino conducted, some testers pointed out that a number of retro objects in the scenes interrupted the players’ immersion because those objects reminded participants of past memories. When designing scenes, therefore, we tried to create

fantasy-like environments instead of real-world-like environments for a more immersive experience.

The other sound-based game is *Dark Echo*, a sound-based horror game [21]. In this game, the main mission is to escape from threatening environments by avoiding scary sounds. This game is acclaimed as a success case of utilizing sounds as intriguing elements. In both *Dark Echo* and our game, players must use sounds to complete missions and progress stories. Because this game is a 2D game, 3D spatial sounds are not used. In *the Melody of Mysterious Stones*, sounds are spatialized with an intention to create a more immersive and interesting experience, which turned out to be successful in the user study.

2.2 Meditation Games

Some applications or video games have made meditation more accessible. *Playne* is one of the meditation games that were successful in motivating players to build a daily practice of mindfulness. Whenever the player completes meditation, a beautiful island grows and transforms. Even though *Playne* was developed as a PC game, some studies including the study of Seabrook et al. state that VR meditation applications could significantly enhance the quality of mindfulness practice [16]. Therefore, we designed this game to be a virtual reality game. Another meditation application that we made reference to is *Mediscape*. This game provides meditation experiences with interactive rhythmic soundscapes [6]. Similar to *Mediscape*, *the Melody of Mysterious Stones* was designed with an intention to make meditation more approachable for novice meditation practitioners. Additionally, sounds are utilized as main elements in both for the creation of a better meditation experience and sense of immersion. However, we believe that spatialized sounds could also be used to entertain the players as Belmega and Boiangiu argued [4]. Therefore, we tried to implement sounds as game elements in *the Melody of Mysterious Stones*.

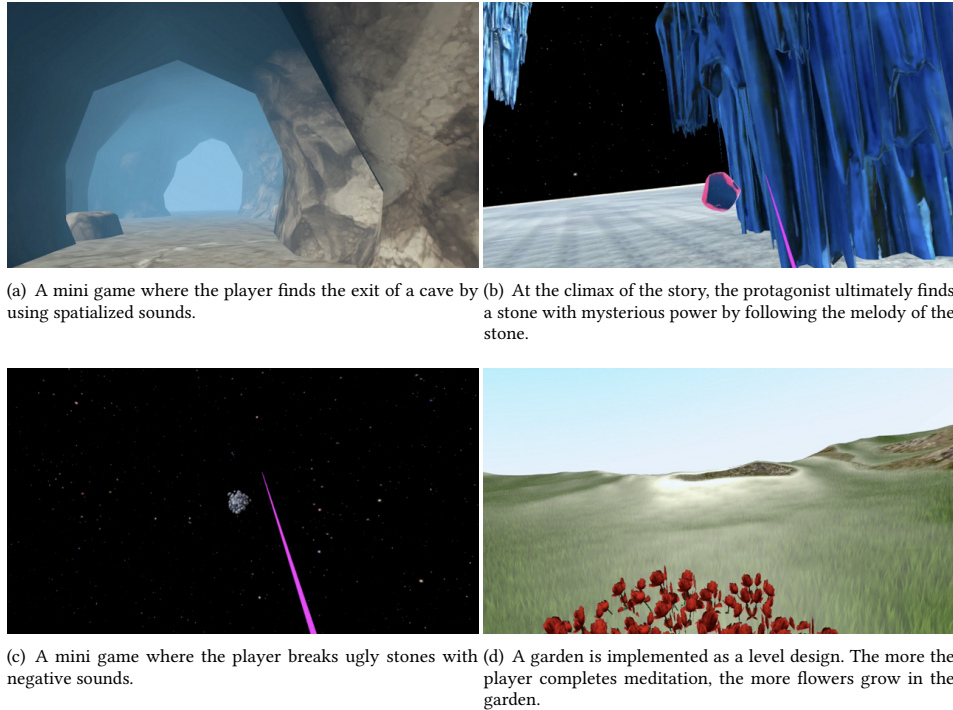


Figure 3: Design of *the Melody of the Mysterious Stones*

3 GAME CONCEPT AND PLAY

The structure of this game, which is described in Figure 2, could be divided into four parts (introduction part, game part, meditation part, and reward part). In the first part, the player must choose her current mood to proceed, and a stone that corresponds to her mood is introduced. In the second part, the player must progress the storyline, complete several missions, and find the stone that is introduced in the first part. In the third part, the player could meditate with materials that vary depending on the player's answer in the first part. Lastly, the player is rewarded after successfully completing meditation. When designing *the Melody of the Mysterious Stones*, we intended it to be enjoyable, encouraging, and immersive as described in Figure 3. For these purposes, we implemented strong storylines, spatialized sounds, mini games, interesting meditation materials, and motivational factors within the context of this structure. The details are explained below.

3.1 Platform

Even though *the Melody of Mysterious Stones* was previously designed as a PC game, it was switched to a VR game to enhance a sense of immersion. Additionally, some studies have revealed that some players do not find traditional video games to be interesting enough to motivate them to play them [3]. Because this game is an Android VR game, a VR device that supports Android such as Oculus Quest 2 is required for playing this game.

3.2 Storyline

The story is in first person. In the town that the protagonist lives in, there is a story that has been passed down from generations to generation. According to the story, there are stones with mysterious powers in the town, and whoever finds all of the stones would reach a state of complete peacefulness. However, nobody in this town knows how many mysterious stones there are, and how many people have found stones. One day, the protagonist walks around a garden without any flowers in the town, and starts to hear beautiful melodies. The protagonist wonders where the sound is coming from, so s/he decides to follow the direction of the sound. The protagonist arrives at a mysterious place that s/he has never been to. In that place, s/he finally finds a mysterious stone. As soon as s/he finds it, the protagonist moves up to space. In space, s/he finds ugly stones with annoying sounds. After breaking them, s/he has a mysterious experience where s/he feels like s/he is free from anxiety and anxiety. This experience is, in the real world, meditation. Once completing meditation, s/he recognizes that s/he has already come back to the town that she lives in, and she finds that some flowers have started to grow in the garden. Afterwards, s/he gets to find stones one by one. Whenever s/he finds one, s/he notices that new flowers begin to grow in the garden.

3.3 Spatialized Sounds

This game uses spatialized sounds for two reasons. One is to intrigue players by providing missions to be completed. The other reason is to help them focus on their aural sense. Even though some studies argue that flow could be negatively correlated with mindfulness,

many studies have proven that they could also have a positive correlation [17]. Specifically, the study of Marty-Dugas shows that paying attention to senses could effectively facilitate flow [12]. Therefore, taking advantage of spatialized sounds in gameplay is expected to assist users in getting the experience of flow as well as achieving a mindfulness state.

3.4 Mini Games

In order to meditate, a player should complete three mini games. These games were implemented not only to entertain players but also to facilitate flow. It has been proposed that flow arises when an individual confronts some challenging situations where the skills of the individual are necessary [13]. Furthermore, some studies have suggested that the experience of flow occurs when the balance between the difficulty of a given challenge and the skills of the individual is retained [18]. Therefore, these games were designed to provide missions at the appropriate levels of difficulties.

3.4.1 Mini Game 1. The first mini game is to find the exit when exploring a cave by using spatialized sounds as a guide. Once finding the exit, the player could move to the location where a mysterious stone is hidden.

3.4.2 Mini Game 2. The second mini game is to find the mysterious stone. Because the melody of the stone would be coming from the direction where the stone is located, the player should use the spatialized sound as a hint in this mini game as well.

3.4.3 Mini Game 3. The last mini game that the player should complete is a stone-breaking mini game where s/he should tear down ugly stones with annoying sounds. Once this game is successfully completed, the player gets meditation experience as a reward.

3.5 Meditation Materials

Once the player completes all of the missions, the players could start meditation as described in Figure 4. The script of audio narration and background music vary depending on the mood that the player chose before the gameplay. Before meditating, a warm-up session is provided. During the session, the player warms up for meditation with enjoying sound spatialization and visualization. During meditating, the player could put down controllers in a comfortable position and listen to audio narration with relaxing soundtracks.

3.6 Motivational Factors

One advantage that gamification brought out is that games are effective in enhancing motivation [7]. Specifically, game contexts that have well-differentiated goal structures could enhance motivation and performance [9]. Therefore, *the Melody of Mysterious Stones* has some motivational factors with hierarchical goal structures that motivate the players to maintain their meditation practice. First of all, the story begins with the protagonist observing a garden without any flowers, but the more meditation activities the player completes, the more flowers with different colors appear in the garden. Additionally, the story describes that the protagonist would reach a state of complete peacefulness once finding all of the stones. This storyline makes the main goal of the game as finding all of the stones. Because the player is able to find different kinds of stones

whenever completing meditation in the game, motivating players to collect stones could work as one of the motivational factors.

4 USER TESTS

We conducted two sets of user tests. The testers responded to two questionnaires. One was a pre-questionnaire that was performed before the gameplay, and the other was done after the gameplay. In these questionnaires, we intended to evaluate the efficiency of spatial audio within the context of the VR meditation game. The result of the first user test revealed the necessity for the revision of the questionnaire as well as test environments and gameplay. The differences between the two are explained and analyzed below.

4.1 Participants

We have 3 male testers and 4 female testers participants in the first user test, and 2 male testers and 3 female testers participated in the second one. Since there were a total of 7 female testers and 5 male testers, it could be said that the gender distribution was even. The age of 3 testers were from 21 to 25, 7 were from 26 to 30, and 2 were from 31 to 35.

4.2 Change of Emotions

In both, a ten-point Likert Scale of 1 (not at all) to 10 (very much) was used to measure the emotion of the testers. Only 33.3% of the testers replied that they have strongly positive feelings by rating it as 8 or more before the gameplay. After the gameplay, however, 81.9% of the testers rated their feeling as 8 or more. This result could be interpreted that the game is effective in improving the mood of the players. However, we observed that one of the testers reported that the intensity of positive emotions has decreased because he felt excited before the gameplay, but felt calmed after the gameplay. To measure the efficiency more correctly in the second user test, we utilized the PANAS (Positive and Negative Affect Schedule) test, one of the most commonly used self-report questionnaires for measuring the change of both positive emotions and negative emotions [8]. By using the PANAS test, 10 positive emotions and 10 negative emotions could be rated on a five-point scale of 1(not at all) to 5(very much). The sum of negative emotion scores before testing was compared to that after testing, and it was observed that the average sum was significantly decreased from 12.4(SD : 6.28) to 4.8(SD : 3.19). Furthermore, the sum of positive emotion scores before testing was compared to that after testing, and it was also observed that the average sum greatly declined from 20.2(SD : 7.96) to 14.2(SD : 5.02). The result could be interpreted that the game affects the intensity of both positive emotions and negative emotions to be decreased because it makes the player feel calmed, which proves that the game was effective in helping the player to achieve mental peace, which is one of the main goals of meditation.

4.3 Immersion and Enjoyableness

The immersion score was rated on a five-point Likert scale of 1(not at all) to 5(very much). The average score in the first test was 2.7 which was lower than expected. To increase the immersion after the first test, the resolution of the meditation video was increased and the testers were guided to watch it in a dark environment. Some additional improvements such as revising the user interface were

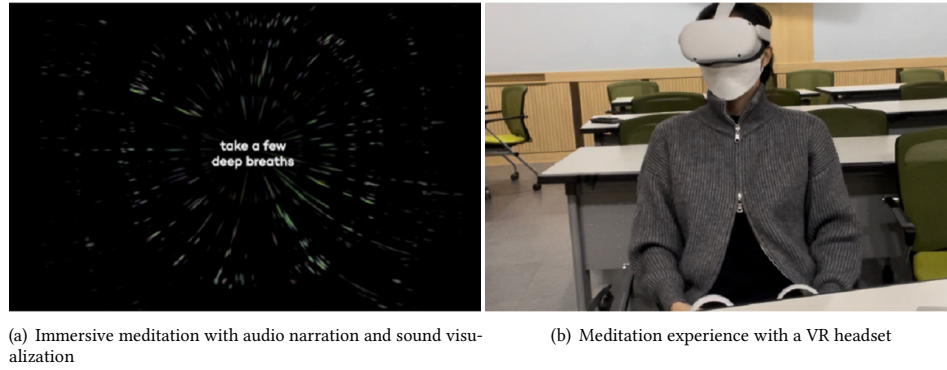


Figure 4: Meditation in *the Melody of the Mysterious Stones*

Table 1: The change of emotion levels after the gameplay

	Before(N=5)	After(N=5)
Positive Effect	Mean: 20.2 SD: 7.96	Mean: 14.2 SD: 5.02
Negative Effect	Mean: 12.4 SD: 6.28	Mean: 4.8 SD: 3.19

Table 2: Comparison on immersion score and enjoyableness score.

	1st Test (N = 7)	2nd Test (N = 5)
Immersion Score	Mean : 2.7 SD : 1.2	Mean : 3.6 SD : 0.9
Enjoyableness Score	Mean : 3.6 SD : 1.4	Mean : 4.2 SD : 0.4

made. As a result, the immersion score has increased from 2.7 to 3.6. The enjoyableness score has also increased from 3.6 to 4.2.

4.4 Meditation Materials

75% of the testers responded that they enjoyed meditation materials in the game, and 41% of them replied that they could concentrate on the materials when practicing meditation.

4.5 Spatial Audio

All of the testers answered that they enjoyed spatial audio elements. Additionally, all replied that spatial audio elements in the game helped them with noticing their aural sense. However, only 67% of the testers replied that they were able to follow the guide of spatialized sounds to complete missions because built-in speakers in Oculus Quest2 did not accurately represent spatial audio.

4.6 Aesthetic Elements

The testers were satisfied with aesthetic elements in the game as well. More than 80% of them replied that visual and sound elements were satisfying enough.

5 CRITICAL REFLECTION

The Melody of the Mysterious Stones was designed to assist those who want to begin practicing meditation. We utilized spatial audio not only to make mindfulness meditation more approachable and enjoyable, but also to help the players focus on their aural sense. All of the testers participating in the preliminary testers replied that they enjoyed using spatial audio as game elements and spatialized sounds helped them notice their sense of hearing. However, the user tests conducted in this research revealed some limitations. Even though the revision after the first test led to an increase in the immersive score and enjoyableness score in the second user test, some users still reported that there were some distracting elements such as visuals. Even though sound visualizations in the meditation materials could be attracting components, they were revealed to be potential distractors from meditation practice. Another limitation of this study is that the test was conducted with a small number of respondents in a short period of time. In this study, a PANAS test was used to observe instantaneous changes after playing this game once. To measure efficiency more correctly, it is needed to conduct a survey with a larger number of participants in a longer period to observe long-term change in emotions and feelings with a more widely-used scale in mindfulness studies such as MAAS (Mindful Attention Awareness Scale) [5]. In addition, because it could be pointed out that the stone-breaking mini game would not be beneficial in calming players, it is needed to be observed if the stone-breaking mini game affects the effectiveness of meditation exercise which would come right after completing this mini game. Further improvement could be made by implementing in-depth level design, for example, by providing level-specific meditation materials and sounds. Such implementation by the development of meditation materials for advanced users is expected to promote long-term use and motivate players to keep up a regular practice.

6 CONCLUSIONS

The Melody of Mysterious Stones is a VR meditation game that takes advantage of spatial audio technologies. It is a sound-based game where sounds are essential elements throughout the gameplay. The main target of this game is those who are interested in mindfulness meditation but have no or little meditation experience. To make the

transition from gameplay to meditation go smoothly and interestingly, several missions where users should use spatialized sounds as important cues have been implemented. Once the missions are successfully completed, the user could enjoy sound spatialization and visualization artwork, and proceed to meditate with audio narration and calming soundtracks. The result of the user test reveals that the users were able to enjoy spatialized sounds as well as get a sense of relaxation through meditation.

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